Vol 7 Chapter 8 — Lagrangian Formulation of EM

Keeper (author pass — deep math/physics revision)

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Chapter 8 — Lagrangian Formulation of Electromagnetism

Level 1 — one sentence

The EM Lagrangian $\mathcal{L}_{EM} = -F_{\mu\nu}F^{\mu\nu}/(4\mu_0) - J^\mu A_\mu$ is the substrate's $U(1)$ Yang-Mills action restricted to the abelian case, and Maxwell's equations are the Euler-Lagrange equations of this Lagrangian with A^μ as field variable — the variational principle is the substrate's natural way of selecting which field configurations the substrate K-types support.

Level 2 — graduate-physicist precision

8.1 The EM Lagrangian density

For the EM field plus a source current J^μ:

$$\mathcal{L}_{EM} = -\frac{1}{4\mu_0}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

In terms of $\vec{E}$ and $\vec{B}$:

$$\mathcal{L}_{EM} = \frac{1}{2}(\epsilon_0 E^2 - B^2/\mu_0) - \rho\phi + \vec{J}\cdot\vec{A}$$

Action: $S = \int d^4x\,\mathcal{L}_{EM}$.

8.2 Maxwell from Euler-Lagrange

Variational principle: $\delta S/\delta A^\nu = 0$.

$$\frac{\partial\mathcal{L}}{\partial A_\nu} - \partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\nu)} = 0$$

Computing: $\partial\mathcal{L}/\partial A_\nu = -J^\nu$. $\partial\mathcal{L}/\partial(\partial_\mu A_\nu) = -(1/\mu_0)F^{\mu\nu}$.

So Euler-Lagrange gives:

$$-J^\nu + \partial_\mu\frac{1}{\mu_0}F^{\mu\nu} = 0 \implies \partial_\mu F^{\mu\nu} = \mu_0 J^\nu$$

The source-equation form of Maxwell's equations.

The other Maxwell equation (Bianchi: $\partial_{[\mu}F_{\nu\rho]} = 0$) is automatic from $F = dA$.

Substrate-mechanism reading: the Lagrangian is the substrate's natural $U(1)$ Yang-Mills action. The variational principle is the substrate's "select stationary configurations" rule.

8.3 Gauge invariance of the action

Under $A^\mu \to A^\mu + \partial^\mu \Lambda$: - $F^{\mu\nu}$ is invariant - $J^\mu A_\mu \to J^\mu A_\mu + J^\mu \partial_\mu \Lambda = J^\mu A_\mu + \partial_\mu (J^\mu \Lambda) - \Lambda \partial_\mu J^\mu$

For conserved current ($\partial_\mu J^\mu = 0$), the action changes by a boundary term, leaving equations of motion invariant. Gauge invariance of EM is consistent with charge conservation (Noether's theorem).

8.4 Stress-energy tensor of EM

From the Lagrangian, the symmetric energy-momentum tensor:

$$T^{\mu\nu}_{EM} = \frac{1}{\mu_0}\left(F^{\mu\alpha}F^{\nu}{}_{\alpha} - \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right)$$

Components: - $T^{00} = u = $ energy density = $(\epsilon_0/2)E^2 + (1/(2\mu_0))B^2$ - $T^{0i} = c\cdot$ Poynting vector $\vec{S} = \vec{E} \times \vec{B}/\mu_0$ - $T^{ij} = $ Maxwell stress tensor (force per area between EM fields)

Conservation: $\partial_\mu T^{\mu\nu}_{EM} = -F^{\nu\lambda}J_\lambda$ — exchange of energy-momentum between field and matter.

8.5 Trace of stress-energy tensor

$T^\mu_\mu = (1/\mu_0)(F^{\mu\alpha}F_{\mu\alpha} - F^{\alpha\beta}F_{\alpha\beta}) = 0$ — the EM stress-energy is traceless. This is a consequence of the conformal invariance of free EM (Section 8.6).

Substrate reading: tracelessness reflects the substrate's $SO(4,2)$ conformal symmetry inherited by the $U(1)$ gauge sector. Conformal invariance is broken only when we add masses (matter Lagrangians); the free EM Lagrangian is conformally invariant.

8.6 Conformal invariance

The free EM Lagrangian (no sources) is invariant under conformal transformations in 4D — the larger group including Lorentz transformations + dilations + special conformal. This is the $SO(4,2)$ structure from Volume 7 Chapter 7.

Substrate reading: conformal invariance of EM is the substrate's natural symmetry of its $U(1)$ gauge sector — the substrate-K-type structure of the photon respects the full conformal group.

8.7 Canonical quantization

Treating A^μ as a field operator with conjugate momenta $\pi^\mu = \partial\mathcal{L}/\partial(\partial_t A_\mu)$: Gauss law constraint, gauge fixing, ghost fields, BRST quantization — the full QED quantization story (Volume 1 Chapter 8 of this curriculum, or Peskin and Schroeder Ch 4-5).

Substrate reading: QED is the quantization of the substrate $U(1)$ gauge field; the photon as quantized excitation of A^μ corresponds to the substrate's Pin(2) integer-weight K-type creation operator.

8.8 K-audit anchors

- Chapter 2: Maxwell from substrate U(1) (Lagrangian foundation)
- Volume 1 Chapter 8: Yang-Mills gauge action (general G; abelian U(1) is the EM case)
- Volume 7 Chapter 7: SO(4,2) conformal symmetry

Level 3 — 5th-grader accessibility

Lagrangian formulation: write down one number, the Lagrangian density $\mathcal{L}$, that depends on the field A^μ and its derivatives. Then find the field configurations that make the integrated action $S = \int \mathcal{L} d^4x$ stationary (Euler-Lagrange equations). For EM: $\mathcal{L} = -F^2/4 - J \cdot A$. Out come Maxwell's equations. Why bother: the variational form is the natural framework for relativistic field theory, gauge invariance, and quantization. In BST: this is the substrate's $U(1)$ Yang-Mills action — the substrate's natural-action form for the abelian gauge sector. Same physics, more elegant math.

What comes next

Chapter 9 develops multipole expansion and scattering.

Where to look this up

- Lagrangian EM: Jackson Ch 12; Goldstein Ch 13
- QED quantization: Peskin and Schroeder Ch 4-5
- BST anchors: Chapter 2; Volume 1 Chapter 8